

Roadmap

This structured, step-by-step roadmap will take you from installation to writing your own programs.

Section 1: Set Up Your Tools

Before writing code, you need an environment to run it

- **Install the JDK:** Download the latest [Oracle Java Development Kit \(JDK\)](#). JDK translates your code into a language your computer understands.
- **Get an IDE:** Download an Integrated Development Environment (IDE) to write your code. include [Microsoft VS Code](#) with Java pack.
- Use the unified "Coding Pack for Java" installer (available for Windows and macOS) to gets **VS Code (IDE), the JDK, and crucial extensions simultaneously**.

Download and install from:

https://code.visualstudio.com/docs/java/java-tutorial#_setting-up-vs-code-for-java-development

Section 2: Learn Core Syntax

Start by learning how a basic program is structured. Every runnable Java program requires a primary entryway called the main method. Your initial checklist should include:

- **Variables:** Containers that hold data, like numbers or text.
 - **Data Representation:** Bit, Byte, and HexDecimal usage.
 - **Data Types:** Differences between integers (int), decimals (double), characters (char), and true/false statements (boolean).
 - **Control Flow:** Using if/else conditions to make decisions.
 - **Loops:** Using for, while loops to repeat tasks automatically.
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Section 3: Understand Object-Oriented Programming (OOP)

Java is fundamentally built on OOP. This means you organize your code into modular, reusable blocks. Master these four pillars:

- **Classes & Objects:** A class is a blueprint, and an object is the actual thing you build from it.
 - **Inheritance:** Letting a new class adopt the properties of an existing class.
 - **Polymorphism:** Allowing different classes to be treated as the same general type.
 - **Encapsulation:** Hiding data inside a class to keep it safe from outside interference.
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Section 4: Mini project

Build a Student Grade & Honor Roll Tracker

Section 5: Git and GitHub basics

Install Git, GitHub desktop, Bash commands and Git commands, Local and remote repository.